Unified Web App for Brain Studies

Objective

To integrate tools of brain studies into a unified web app for convenient access.

Overall

Access from the lab's local network via http://133.68.119.20:7777.
The web app (***Fig. 1***) unifies four tools — isocond, anicond, mesh_head, and mesh_sim — covering conductivity mapping (isotropic and anisotropic), head mesh generation, and TMS simulation.

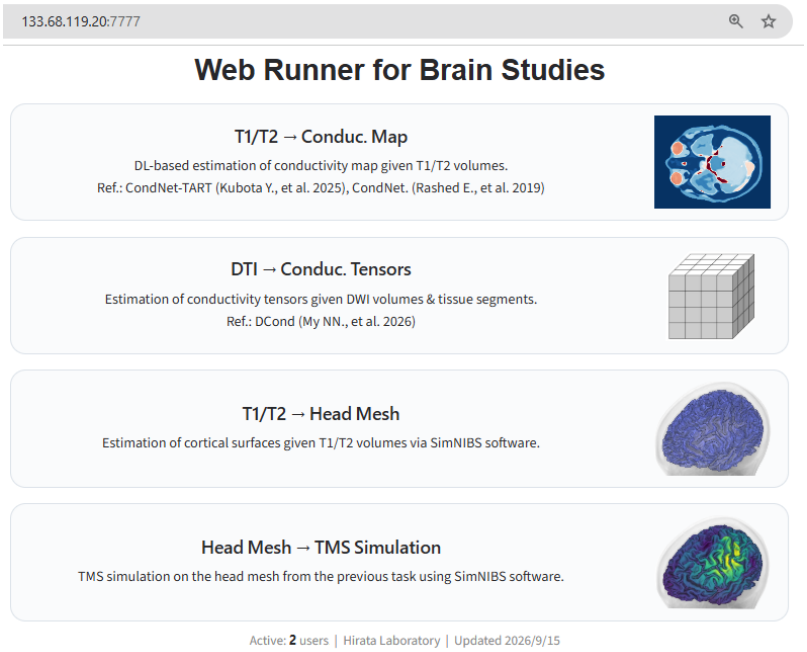


Fig. 1. Web runner homepage

Tool 1. T1/T2 → Conductivity Map (isocond)

- Referred from studies of Kubota-sensei and Prof. Rashed of the lab.
- Same as the model runner mentioned in the previous email on 4th August, 2026 (***Fig. 2***).

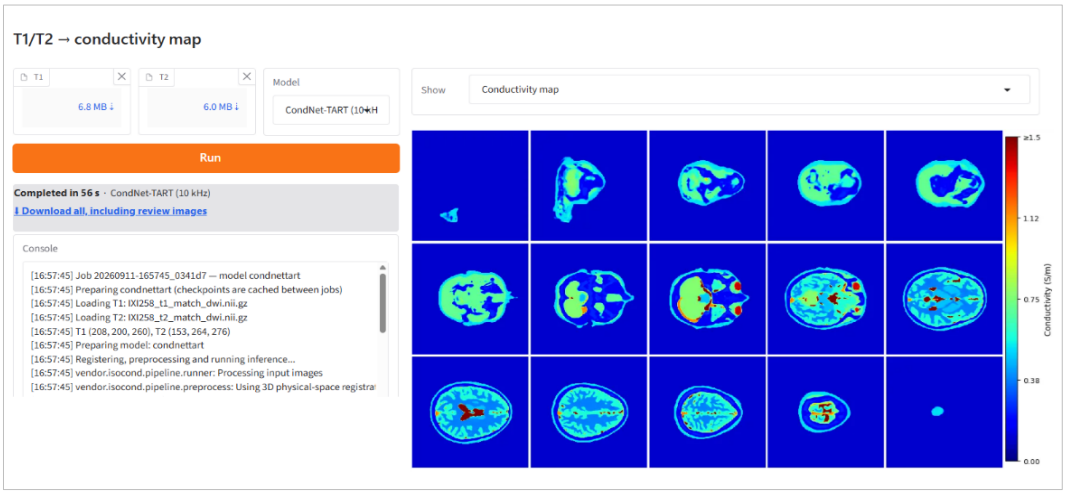


Fig. 2. isocond tool for estimating the isotropic conductivity map

Tool 2. DTI → Conductivity Tensors (anicond)

- This tool is to implement the on-going study about estimating the anisotropic conductivity from brain diffusion MRI (***Fig. 3***). Three models are included: direct-mapping/DM (conventional), volume-normalized/VN (conventional), and tract-based/TBe (original).
- **Inputs.** The DWI series (one 4D file or the per-direction volumes) as NIfTI, its bval and bvec files, and a tissue segmentation. Optional: the conductivity table (the shipped default can be downloaded, edited and re-uploaded), and a whole-brain tractogram.
- **Outputs.** The conductivity tensor as a 6-component NIfTI (xx, xy, xz, yy, yz, zz), which is the tensor Tool 4 accepts. Intermediates: the anisotropic mask, the diffusion tensor, the whole-brain tractogram (TBe only), the effective conductivity map and the eigenvalue histogram within the anisotropic regions (***Fig. 4***).
- **Timing.** DM/VN take seconds. TBe takes several minutes (faster if tractogram uploaded).

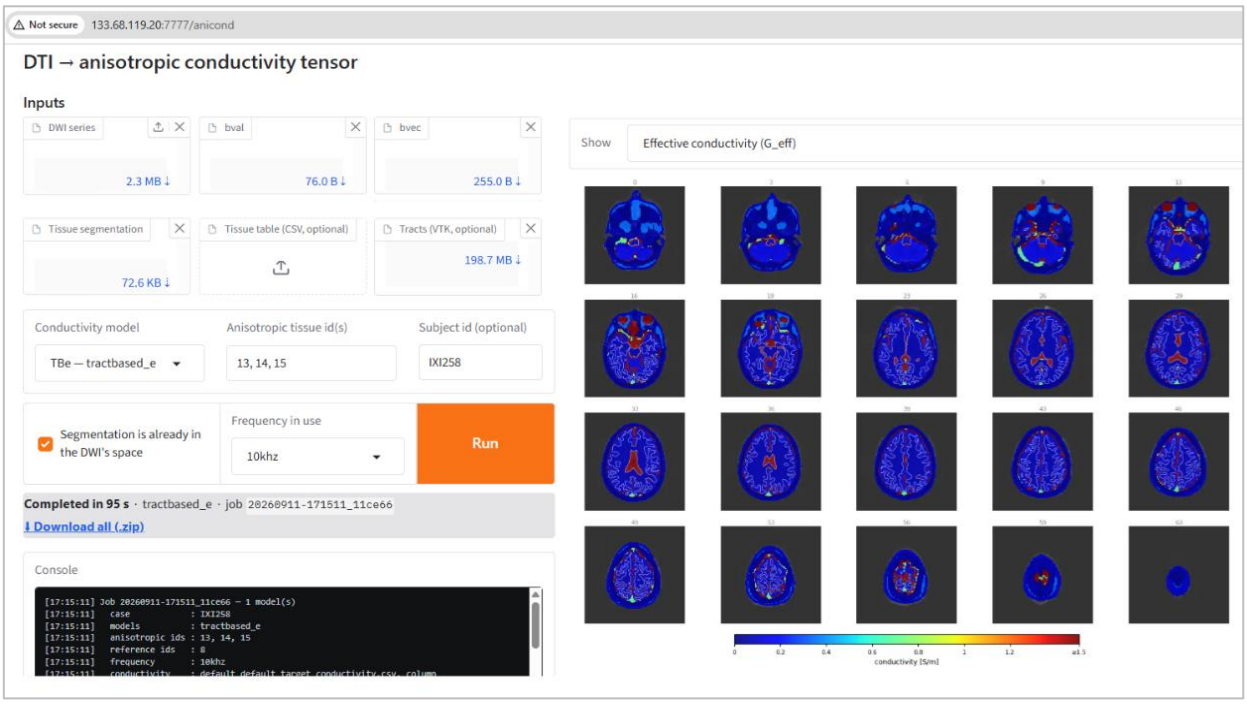


Fig. 3. anicond tool for estimating the anisotropic conductivity tensors

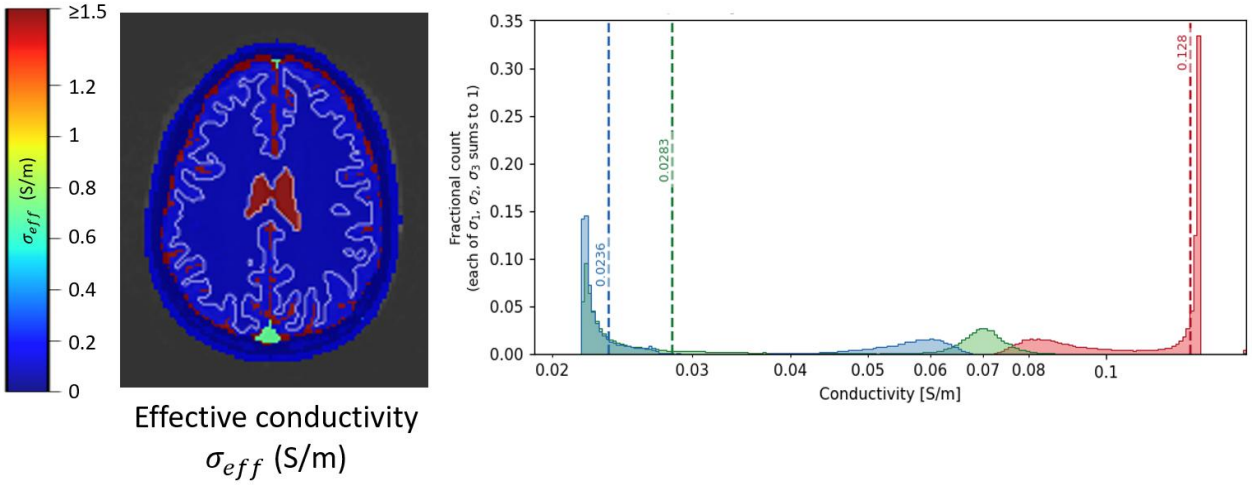


Fig. 4. Some anisotropic conductivity outputs returned by the anicond tool

Tool 3. T1/T2 → Head Mesh (mesh_head)

- This tool is to estimate tissue segments and cortical surfaces from T1 (and T2) images via SimNIBS software functions (***Fig. 5***).
- **Inputs.** A T1 and a T2 (optional) as NIfTI. If the images are not in the DTI's space, the b0 image is uploaded as well and the T1 is registered to it rigidly. This ensures the tensor from Tool 2 and the simulation in Tool 4 overlay correctly.
- **Outputs.** An m2m folder: the head mesh (.msh, 9 tissues), the tissue segmentation, the white, central and pial surfaces (***Fig. 6***), and EEG electrode positions. Everything is zipped (about 700 MB); this zip is the input of Tool 4.
- **Timing.** Up to 60 minutes per subject.

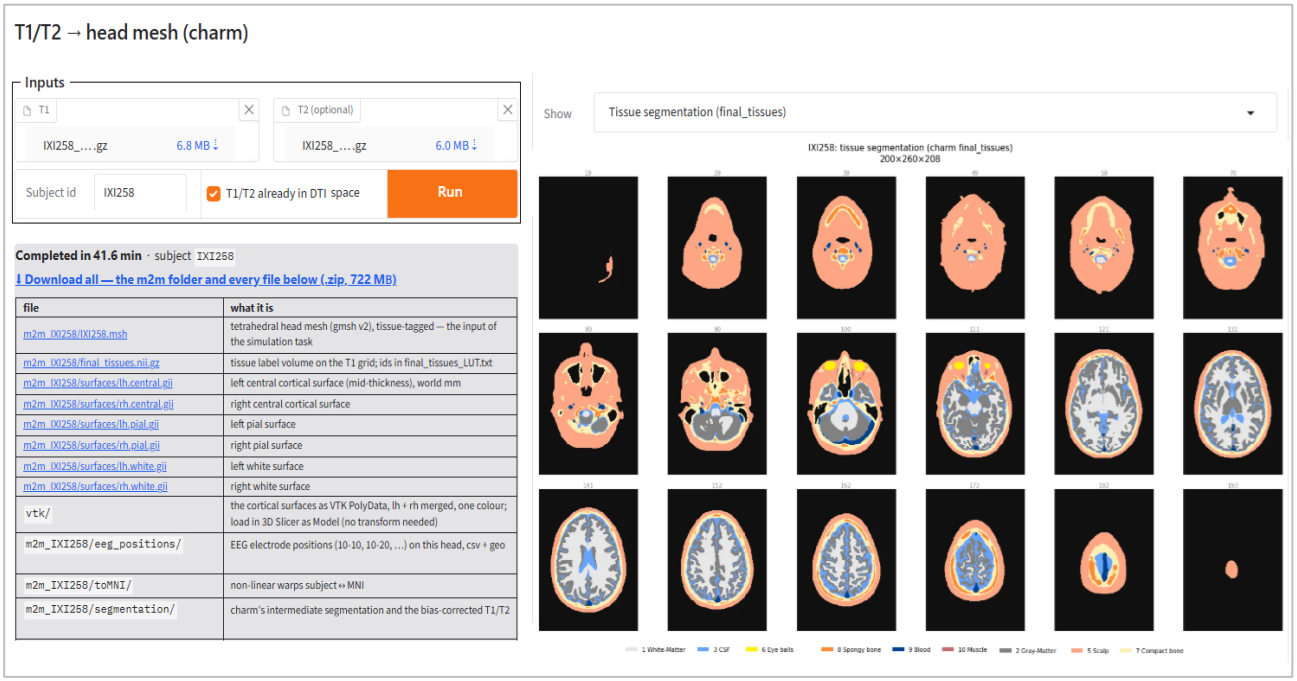


Fig. 5. mesh_head tool for estimating tissue segments and cortical surfaces

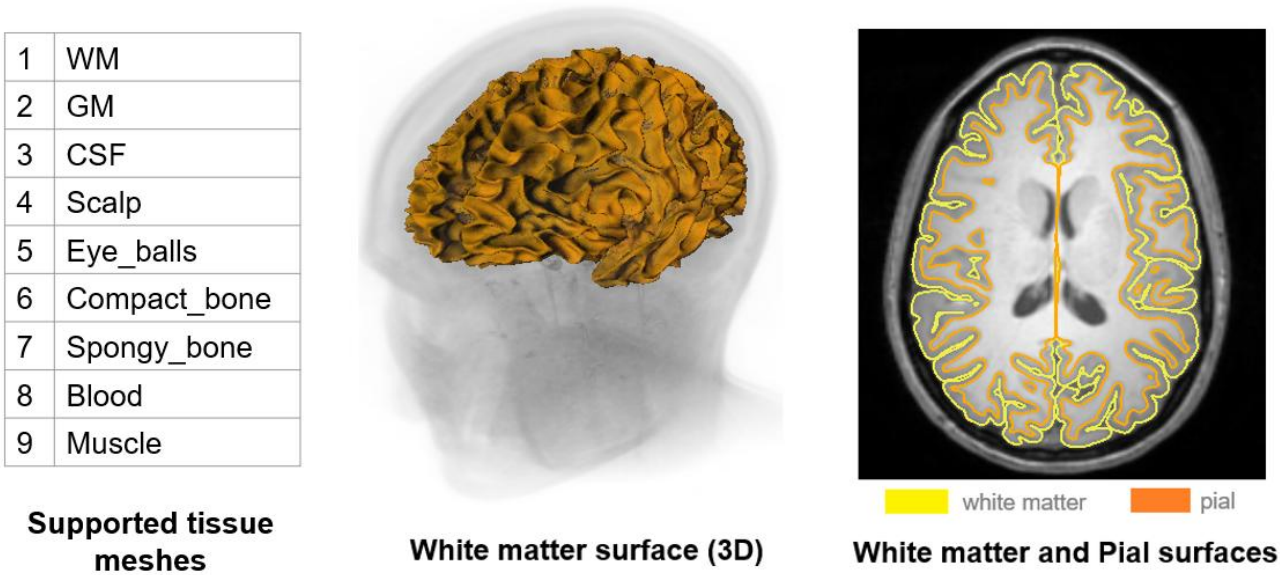


Fig. 6. Output objects generated by the mesh_head tool

Tool 4. Head Mesh → TMS Simulation (mesh_sim)

Run SimNIBS's FEM solver for the head model from Tool 3 (***Fig. 7***).

Inputs. The zip from Tool 3. Coil placement, coil model, and dI/dt. dI/dt is capped at the selected coil's measured maximum at 100% stimulator output (1.48e8 A/s for the MagVenture MCF-B65). For an anisotropic run, the tensor from Tool 2 is uploaded.

Outputs (***Fig. 8***). The induced E-field on the whole head mesh (.msh), on the cortical surfaces. Review images: the field on the GM and WM, the conductivity tensor's principal direction, and the ICNIRP exceedance map, i.e. |E| divided by the ICNIRP 2010 occupational restriction at 10 kHz (3.82 V/m peak), drawn where the ratio exceeds 1.

Time. several minutes per coil position.

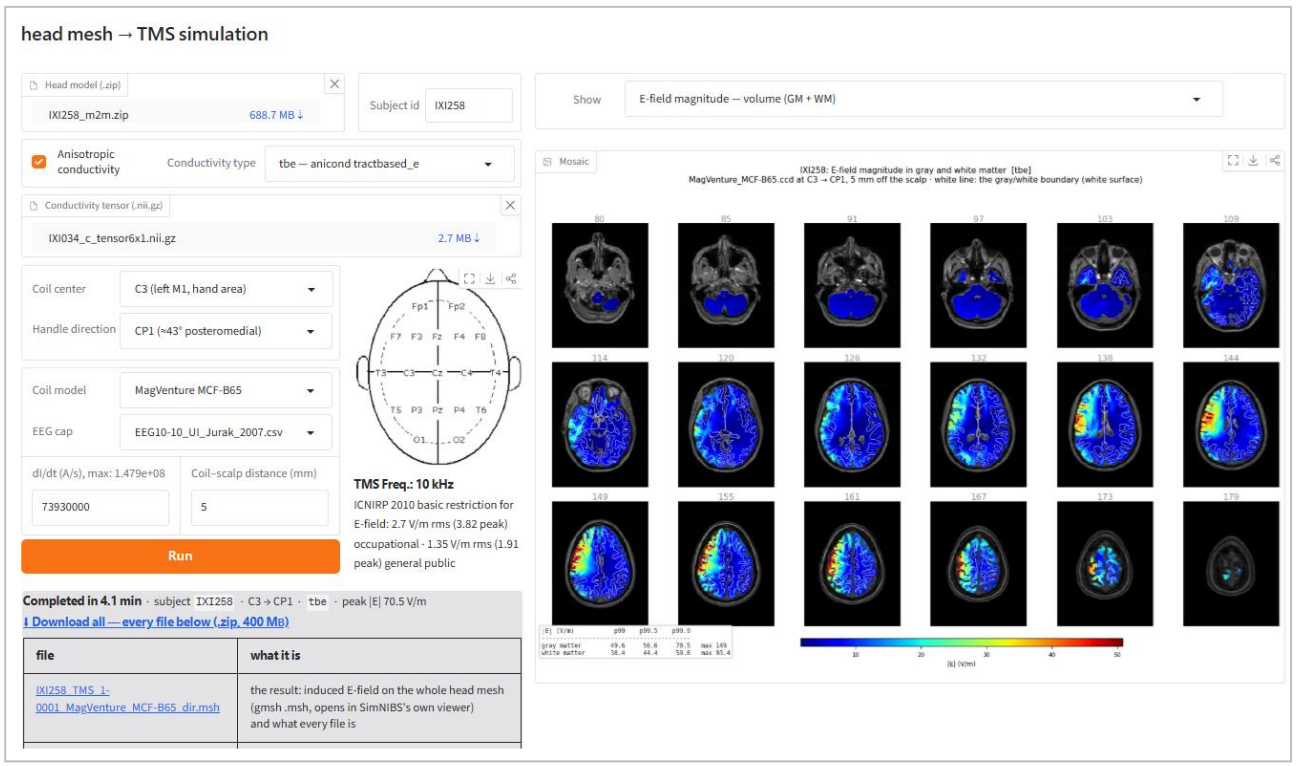
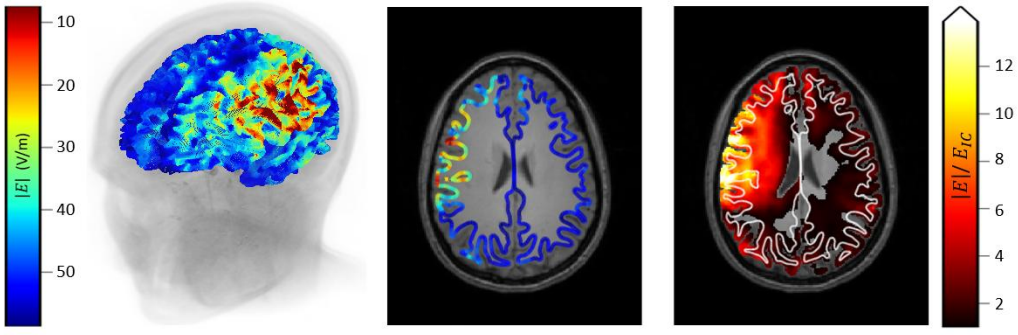


Fig. 7. mesh_sim tool for performing TMS simulation



|E| (V/m)	p99	p99.5	p99.9
gray matter	49.6	56.6	70.5
white matter	38.4	44.4	59.8

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ICNIRP 2010 occup., peak, 10 kHz:  3.82 V/m
GM+WM above it: 75.7% of the volume
max ratio 28.3   dI/dt 7.393e+07 A/s
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s

Fig. 8. Induced E-field and related outputs returned by the mesh_sim tool